

Monster

Jojo likes to play Regnerok Online, he have a list of N monster that he can kill. Jojo is not really good at math. Now he asks you which monster he should kill. Jojo wants the level difference between the monster and Jojo to be as minimum as possible.

Format Input

The first line of the input will be T, the number of the test cases ($1 \leq T \leq 10$).

Each test case starts with integers N and M, number of the monster in the list and Jojo's level.

The next line of the input will contain N integers, the i-th integer L_i will describe the level of the i-th monster.

It is guaranteed that the monster levels will be between 1 and 1000000.

Format Output

Output in one line "Case #T: " followed by the index of the monster that Jojo should kill. If there are multiple answer, print the one with the lowest index.

Constraints

$1 \leq T \leq 10$

$1 \leq N \leq 1000000$

$1 \leq M, L_i \leq 1000000$

Sample Input 1	Sample Output 1
3 5 5 1 2 3 4 5 5 5 5 5 5 5 5 5 10 1 2 3 4 5	Case #1: 5 Case #2: 1 Case #3: 5

Sample Input 2	Sample Output 2
3 10 5 1 2 3 4 5 6 7 8 9 10 5 10 10 20 30 40 50 3 5 100 100 100	Case #1: 5 Case #2: 1 Case #3: 1